Abdulaziz University
Faculty of Computing & Information Technology
Computer Science Department

SOLUTION

Mid Term Exam (Make Up)

Thursday October 12, 2023

Course: CPCS223 (Analysis & Design of Algorithms)

Term: Fall 2023 (1st Semester)

Time: 80 minutes

Student ID: ______________________________ **Section:** CS1 / CS2

Student Name: ______________________________

Questions	Marks	Obtained Marks
1	10	
2	15	
3	5	
4	20	
5	10	
Total Marks	**60**	

Instructors:

- Dr. Muhammad Umair

<u>Question No. 1 (SO-2, 10 Marks)</u>

Part – A

Write down at least 5 characteristics of a good algorithms

<u>Solution:</u>

Question 01

Which of the following is / are not the basic characteristic of an **Algorithm?**

a) Step by Step Method
b) Unambiguous Instruction
c) Computational Procedure
d) Instances

Question 02

The running time of "Selection Sort" is:

a) $O(n^2)$
b) O(n)
c) O(n lg n)
d) O(lg n)

Question 03

For RAM model, which one is not the basic assumption of the model?

a) Infinitely large random-access memory
b) Every instruction completes its execution in "unit" time
c) Instructions do not execute sequentially
d) Reproducible execution time

Question 04

What will be the value of $\sum_{k=0}^{n} m$

a) n * *m*
b) n
c) n + m
d) (n + 1) * *m*

Question 05

According to the definition, which condition(s) is / are correct for Theta-Notation?

a) f(n) > 0
b) c_1 and c_2 must be positive integers
c) n can be negative
d) It describes Upper Bound

Part – B

Perform the "Worst Case Analysis" of the given algorithm, using RAM model

COST **Times**

```
INSERTION-SORT(A)
for j ← 2 to length[A]
     do key ← A[j]
        ▷ Insert A[j] into the sorted
                sequence A[1 . . j − 1].
        i ← j − 1
        while i > 0 and A[i] > key
            do A[i + 1] ← A[i]
               i ← i − 1
        A[i + 1] ← key
```

<u>Solution:</u>

<u>Part – B</u>
Perform the "Worst Case Analysis" of the given algorithm, using RAM model

<u>Question No. 2</u> **<u>(SO-2, 15 Mdarks)</u>**

Design a recursive algorithm for computing 2^n for any nonnegative integer n that is based on the formula:

$$2^n = 2^{n-1} + 2^{n-1}.$$

Then give the corresponding complexity using **"Backward Substitution"** method

Note: the design can be provided as a pseudocode or as explicit and clear steps.

<u>Solution:</u>

Algorithm:

(05 pts)

```
Tow_Powern(n)
{
        If n=0 return (1)
        Else return Tow_Powern(n-1) + Tow_Powern(n-1)
}
```

The corresponding complexity is 2^n

Recurrence

$T(n) = T(n-1) + T(n-1) + 1$

1

$T(n) = 2\ T(n-1) + 1$

Backward Substitution

By backward substitution

$n = n - 1$ **$T(n-1) = 2\ T(n-2) + 1$**

2 **$T(n) = 2\ [2\ T(n-2) + 1\ ] + 1$** **$T(n) = 4\ T(n-2) + 3$**

$n = n - 2$ **$T(n-2) = 2\ T(n - 3) + 1$**

3 **$T(n) = 4\ [2\ T(n - 3) + 1] + 3$** **$T(n) = 8\ T(n-3) + 7$**

$n = n-3$ **$T(n-3) = 2\ T(n-4) + 1$**

4 **$T(n) = 8\ [2\ T(n-4) + 1] + 7$** **$T(n) = 16\ T(n-4) + 15$**

(05 pts)

So for "i th " call

$$T(n) = 2^i\ T(n - i\ ) + 2^i - 1$$

(03 pts)

This recursion will stop as $n - i = 0$, that implies $n = i$

$$
\begin{aligned}
T(n) &= 2^n\ T(n - n\ ) + 2^n - 1 \\
&= 2^n\ T(0) + 2^n - 1 \\
&= 2^n . 1 + 2^n - 1 \\
&= 2^n + 2^n - 1 \\
&= 2^n
\end{aligned}
$$

(02 pts)

Question No. 3 **(SO-2, 05 Marks)**

Prove / disprove that *f(n) = θ g(n)* for the given function

$$f(n) = an^2 + bn + c$$

Solution:

$a\,n^2 + b\,n + c = \Theta(n^2)$

$f(n) = a\,n^2 + b\,n + c$

and

$g(n) = \Theta(n^2)$

we have to prove

$c2\ g(n) <= f(n) <= c1\ g(n)$

L. H. S (2.5 pts)	R. H. S (2.5 pts)
$c2\,(n^2) <= a\,n^2 + b\,n + c$	$a\,n^2 + b\,n + c <= c1\,n^2$
$c2 <= a + b/n + c/n^2$	$a + b/n + c/n^2 <= c1$
for $n = \infty$	for $n = 1$
this expression is true for all $n >= 1$	this expression is true for all $n >= 1$
and	and
$c2 <= a$	$c1 >= a + b + c$

Question No. 4 **(SO-2, 20 Marks)**

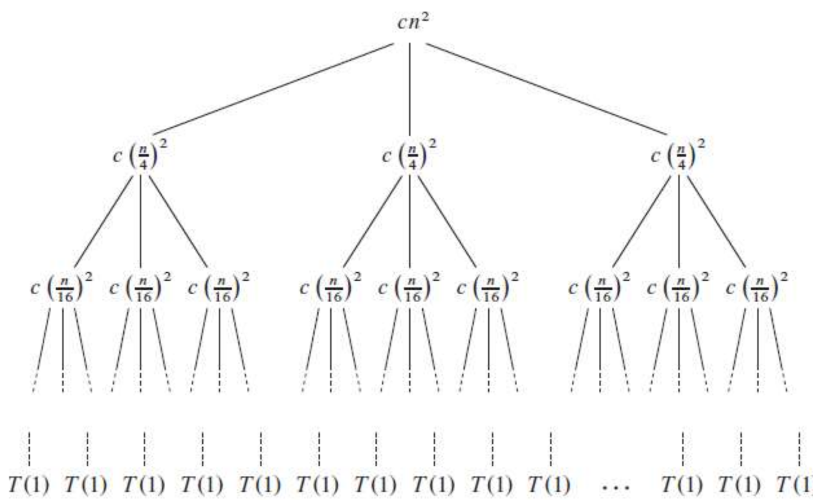
Using the "*Recursion Tree Method*" find the complexity class for the given algorithm. Provide the solution for maximum or minimum running time for the given recurrence.

$$T(n) = 3T(\lfloor n/4 \rfloor) + \Theta(n^2)$$

Also prove your solution using "Substitution Method"

Solution:

Recursion Tree (05 pts)



Problem Size at level "i"	$n/4^i$
Cost of node at level "i"	$c\, n^2/16^i$
No. of nodes at level "i"	3^i
Cost of level "i"	$(3/16)^i\, cn^2$
Last Level number	$\log_4 n$
No. of nodes at last Level	$n^{\log_4 3}$
Cost of last Level	$\Theta(n^{\log_4 3})$

Question No. 4 **(SO-2, 20 Marks)**

Running Time (10 pts)

$$
\begin{aligned}
T(n) &= cn^2 + \frac{3}{16}cn^2 + \left(\frac{3}{16}\right)^2 cn^2 + \cdots + \left(\frac{3}{16}\right)^{\log_4 n - 1} cn^2 + \Theta(n^{\log_4 3}) \\
&= \sum_{i=0}^{\log_4 n - 1} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3}) \\
&= \frac{(3/16)^{\log_4 n} - 1}{(3/16) - 1} cn^2 + \Theta(n^{\log_4 3}) \\
&< \sum_{i=0}^{\infty} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3}) \\
&= \frac{1}{1 - (3/16)} cn^2 + \Theta(n^{\log_4 3}) \\
&= \frac{16}{13} cn^2 + \Theta(n^{\log_4 3}) \qquad = \quad O(n^2)
\end{aligned}
$$

Proof using "Substitution Method" (05pts)

$$
\begin{aligned}
T(n) &\le 3T(\lfloor n/4 \rfloor) + cn^2 \\
&\le 3d\,\lfloor n/4 \rfloor^2 + cn^2 \\
&\le 3d(n/4)^2 + cn^2 \\
&= \frac{3}{16}dn^2 + cn^2 \\
&\le dn^2 ,
\end{aligned}
$$

where the last step holds as long as $d \ge (16/13)c$.

Question No. 5 **(SO-2, 10 Marks)**

Solve the following recurrence relations.

$$x(n) = x(n-1) + 5 \text{ for } n > 1,\ x(1) = 0$$

$$x(n) = x(n/3) + 1 \text{ for } n > 1,\ x(1) = 1$$

Solution:

$x(n) = x(n-1) + 5$ *for* $n > 1$, $x(1) = 0$

a) $x(n) = x(n-1) + 5 \quad \text{for } n > 1, \quad x(1) = 0$

$$\begin{aligned} x(n) &= x(n-1) + 5 \\ &= [x(n-2) + 5] + 5 = x(n-2) + 5 \cdot 2 \\ &= [x(n-3) + 5] + 5 \cdot 2 = x(n-3) + 5 \cdot 3 \\ &= \ldots \\ &= x(n-i) + 5 \cdot i \\ &= \ldots \\ &= x(1) + 5 \cdot (n-1) = 5(n-1). \end{aligned}$$

Note: The solution can also be obtained by using the formula for the n term of the arithmetical progression:

$$x(n) = x(1) + d(n-1) = 0 + 5(n-1) = 5(n-1).$$

Solution:

x (n) = x(n/3) + 1 for n > 1, x(1) = 1

$x(n) = x(n/3) + 1 \quad \text{for } n > 1, \quad x(1) = 1 \quad (\text{solve for } n = 3^k)$

$$\begin{aligned}
x(3^k) &= x(3^{k-1}) + 1 \\
&= [x(3^{k-2}) + 1] + 1 = x(3^{k-2}) + 2 \\
&= [x(3^{k-3}) + 1] + 2 = x(3^{k-3}) + 3 \\
&= \ldots \\
&= x(3^{k-i}) + i \\
&= \ldots \\
&= x(3^{k-k}) + k = x(1) + k = 1 + \log_3 n.
\end{aligned}$$

Extra Page- If you like any work on this page graded, please label your work clearly.